AnDet Pro on Google Colab

This notebook runs the AnDet Pro experiment on Google Colab and saves the results in Google Drive. Run the cells from top to bottom. To run a cell, click it and press Shift + Enter.

Before you start

1. In Google Drive, create a folder named `AnDet`.
2. Put these two files in that folder: `AnDet_Pro_v3_rebuilt.zip` and `NF-CSE-CIC-IDS2018.parquet`.
3. In the Colab menu go to Runtime, then Change runtime type, choose T4 GPU and click Save. The GPU makes training faster.

Step 1: Connect Google Drive and unpack the project

This cell connects Google Drive, checks that the two files are in the AnDet folder, unzips the project, and copies the data file to Colab's own disk because it reads faster there. It ends by printing the project folder, `/content/AnDet_Pro`.

```
from google.colab import drive
drive.mount('/content/drive')

import os, shutil, zipfile
SRC = '/content/drive/MyDrive/AnDet'
DATA = f'{SRC}/NF-CSE-CIC-IDS2018.parquet'
RESULTS = f'{SRC}/results'
assert os.path.exists(f'{SRC}/AnDet_Pro_v3_rebuilt.zip'), 'Put AnDet_Pro_v3_rebuilt.zip in the Drive folder A
assert os.path.exists(DATA), 'Put NF-CSE-CIC-IDS2018.parquet in the Drive folder AnDet'
if not os.path.exists('/content/AnDet_Pro'):
    zipfile.ZipFile(f'{SRC}/AnDet_Pro_v3_rebuilt.zip').extractall('/content')
shutil.copy(DATA, '/content/NF-CSE-CIC-IDS2018.parquet')   # a local copy reads faster than Drive
DATA_LOCAL = '/content/NF-CSE-CIC-IDS2018.parquet'
os.makedirs(RESULTS, exist_ok=True)
%cd /content/AnDet_Pro
```

```
Mounted at /content/drive
/content/AnDet_Pro
```

Step 2: Install pytest and check the set-up

Colab already has TensorFlow, NumPy, pandas, SciPy, PyArrow and Matplotlib, so only pytest needs to be installed. Do not run `pip install -r requirements.txt` on Colab, because it can replace the TensorFlow that Colab already has.

The last line of the output must say **READY**. A note about the GPU is fine.

```
!pip -q install pytest
!python check_setup.py --dataset {DATA_LOCAL}
```

```
AnDet Pro - set-up check
------------------------------------------------------------
[NOTE] Python 3.13.16 on Linux x86_64
       -> TensorFlow may not have a build for this Python yet. Python 3.11 is the safe choice.
[PASS] numpy 2.1.3
[PASS] pandas 2.2.3
[PASS] scipy 1.16.3
[PASS] pyarrow 23.0.1
[PASS] matplotlib 3.10.0
[PASS] pytest 8.4.2
[PASS] TensorFlow 2.21.0 (loaded in 3.9 s)
[PASS] GPU found: /physical_device:GPU:0
[PASS] Data file NF-CSE-CIC-IDS2018.parquet: 62 MB, 5,918,803 rows
[PASS] Label column: Label; attack-type column: not found; 10 features
       features: L4_SRC_PORT, L4_DST_PORT, PROTOCOL, L7_PROTO, IN_BYTES, OUT_BYTES, IN_PKTS, OUT_PKTS, TCP_F
[PASS] config/config.ini is valid
       what each mode runs:
       mode       every n-th flow  max epochs  trials per condition  models trained
       fast                   590           4                    30               1
       standard                59         100                   300               2
       full                    59         100                   300               9
[PASS] Results go to /content/AnDet_Pro/results (75 GB free)
[PASS] VMD test: 4 features x 300 steps -> 5 modes each; the modes add back to the signal within 9.2% (0.09
[PASS] Model test: 250,501 parameters (the thesis says 250,501); 1 epoch on 200 windows took 8.5 s
[PASS] Policy engine test: rules with priorities [1100, 1200, 65000]
------------------------------------------------------------
```

```
15 passed, 1 notes, 0 failed

READY. Next:
    python run.py --mode fast        (a few minutes: checks the whole program)
    python run.py --mode standard    (the run for the thesis)
```

Step 3: Quick test run

This is a small run that takes a few minutes. It only checks that the whole program works. Do not use its numbers in the thesis or the slides.

When it finishes, the output shows `THIS WAS A FAST CHECK`. If you see `SOMETHING WENT WRONG`, stop and fix that first.

```
!PYTHONUNBUFFERED=1 python run.py --mode fast --dataset {DATA_LOCAL} --results-dir {RESULTS}
```

```
--- 2. Build features and train the main model: 15.8 s

=== 3. Time needed to score one window (NFR-05) ===
  one window: VMD 16.2 ms + model 5.9 ms = 22.4 ms (median), 57.0 ms (95th percentile); batch scoring 3006
--- 3. Time needed to score one window (NFR-05): 1.1 s

=== 4. Replay the real test stream (no injected attacks) ===
  AUROC 0.988 (baseline 0.948); at the threshold: recall 0.889, false alarm rate 0.050
--- 4. Replay the real test stream (no injected attacks): 0.1 s

=== 5. Head-to-head trials: AnDet Pro against the static baseline ===
  6 scenarios x 3 repetitions = 18 trials per condition
  AnDet Pro : detected 88.9%, time to detect 23.9 s, false alarms 16.3%, contained 88.9%
  Baseline  : detected 94.4%, time to detect 28.5 s, false alarms 21.3%, contained 88.9%
  look-ahead check: false alarms before the attack 14.3% (VMD on the whole stream) against 12.6% (VMD on pa
--- 5. Head-to-head trials: AnDet Pro against the static baseline: 2.7 s

=== 6. Threshold sweep ===
--- 6. Threshold sweep: 0.2 s

=== 7. Extra checks (none selected) ===
--- 7. Extra checks (none selected): 0.0 s

=== 8. Writing tables, charts and records ===

Results written to /content/drive/MyDrive/AnDet/results/20261029_021510_fast

================================================================================
RESULT SUMMARY   mode: fast   trials per condition: 18   seed: 42
================================================================================
                                        Static baseline           AnDet Pro
Attacks detected (%)                               94.4                 88.9
Time to detect, mean (SD), s                 28.47 (26.04)       23.90 (26.70)
Time to respond, mean (SD), s                66.42 (28.20)       23.90 (26.70)
False positive rate, mean (SD), %            21.35 (32.53)       16.26 (34.01)
Attacks contained (%)                              88.9                 88.9
--------------------------------------------------------------------------------
MTTD   baseline minus AnDet Pro: t(30.8) = 0.50, p = 0.622, d = 0.17 (negligible)
MTTR   baseline minus AnDet Pro: t(31.0) = 4.45, p < 0.001, d = 1.55 (large)
FPR    baseline minus AnDet Pro: t(33.9) = 0.46, p = 0.649, d = 0.15 (negligible)
Containment, two-proportion test: z = 0.00, p = 1.000
--------------------------------------------------------------------------------
Requirements: 3 met, 3 missed, 4 not measured
One window scored in 22.4 ms (median); target 50 ms
Real test stream: AUROC AnDet Pro 0.988, baseline 0.948; normal windows flagged at the threshold: 5.0%
Trials in which at least half of the normal windows were flagged: AnDet Pro 16.7%, baseline 16.7%
********************************************************************************
THIS WAS A FAST CHECK. Do not use these numbers in the thesis or the slides.
********************************************************************************

Total time: 0.5 minutes

WHAT TO DO NEXT
  1. Open  /content/drive/MyDrive/AnDet/results/20261029_021510_fast/RESULTS_FOR_THESIS.md   (tables and d
  2. Open  /content/drive/MyDrive/AnDet/results/20261029_021510_fast/DOCUMENT_UPDATE_SHEET.md   (every plac
  3. Charts are in  /content/drive/MyDrive/AnDet/results/20261029_021510_fast/figures   and every table as
  This was only a check. Now run:   python run.py --mode standard
```

Step 4: The real run

This run gives the numbers for the thesis and the slides.

Training is limited to about 60 minutes (`model.max_minutes=60`). When the limit is reached the best epoch so far is kept, and the results file says so. If this happens, mention it in the thesis. After the first epoch the output shows how many seconds one

epoch takes, so you can see roughly how long the run will last.

Keep this browser tab open until the run ends. The run is finished when the output shows `Total time` and `WHAT TO DO NEXT`, and the summary box says `mode: standard`.

```
!PYTHONUNBUFFERED=1 python run.py --mode standard --dataset {DATA_LOCAL} --results-dir {RESULTS} --set mode
```

```
  epoch  14/100  loss 2.212e-02  val_loss 1.930e-02  lr 1.0e-03  (14.9 s)
  epoch  15/100  loss 2.162e-02  val_loss 1.920e-02  lr 1.0e-03  (15.0 s)
  epoch  16/100  loss 2.142e-02  val_loss 1.871e-02  lr 1.0e-03  (14.9 s)
  epoch  17/100  loss 2.169e-02  val_loss 1.887e-02  lr 1.0e-03  (14.9 s)
  epoch  18/100  loss 2.155e-02  val_loss 1.994e-02  lr 1.0e-03  (14.8 s)
  epoch  19/100  loss 2.165e-02  val_loss 1.917e-02  lr 1.0e-03  (15.0 s)
  epoch  20/100  loss 2.082e-02  val_loss 1.991e-02  lr 1.0e-03  (14.9 s)
  epoch  21/100  loss 2.041e-02  val_loss 1.751e-02  lr 1.0e-03  (14.9 s)
  epoch  22/100  loss 2.000e-02  val_loss 1.915e-02  lr 1.0e-03  (14.9 s)
  epoch  23/100  loss 2.049e-02  val_loss 1.841e-02  lr 1.0e-03  (15.0 s)
  epoch  24/100  loss 2.051e-02  val_loss 1.823e-02  lr 1.0e-03  (15.0 s)
  epoch  25/100  loss 2.134e-02  val_loss 1.947e-02  lr 1.0e-03  (20.6 s)
  epoch  26/100  loss 2.012e-02  val_loss 1.706e-02  lr 1.0e-03  (15.0 s)
  epoch  27/100  loss 2.154e-02  val_loss 1.887e-02  lr 1.0e-03  (14.9 s)
  epoch  28/100  loss 2.104e-02  val_loss 1.759e-02  lr 1.0e-03  (14.9 s)
  epoch  29/100  loss 2.102e-02  val_loss 1.855e-02  lr 1.0e-03  (15.0 s)
  epoch  30/100  loss 2.381e-02  val_loss 2.211e-02  lr 1.0e-03  (15.2 s)
  epoch  31/100  loss 2.368e-02  val_loss 2.063e-02  lr 1.0e-03  (14.9 s)
  epoch  32/100  loss 2.247e-02  val_loss 1.985e-02  lr 5.0e-04  (15.0 s)
  epoch  33/100  loss 2.200e-02  val_loss 1.935e-02  lr 5.0e-04  (15.0 s)
  epoch  34/100  loss 2.154e-02  val_loss 1.902e-02  lr 5.0e-04  (14.9 s)
  epoch  35/100  loss 2.107e-02  val_loss 1.858e-02  lr 5.0e-04  (14.9 s)
  epoch  36/100  loss 2.064e-02  val_loss 1.795e-02  lr 5.0e-04  (14.9 s)
    trained 36 epochs (best 26), loss 2.012e-02, val 1.706e-02; threshold: mu 1.706e-02, sigma 4.502e-03, t
  [no_vmd] AUROC 0.839  trials: detected 92.7%, false alarms 15.6%
--- 7. Extra checks (1 more models): 561.4 s

=== 8. Writing tables, charts and records ===

Results written to /content/drive/MyDrive/AnDet/results/20261029_021641_standard

============================================================================
RESULT SUMMARY   mode: standard   trials per condition: 300   seed: 42
============================================================================
                                        Static baseline           AnDet Pro
Attacks detected (%)                               83.7                78.3
Time to detect, mean (SD), s                 51.53 (64.63)       37.00 (35.31)
Time to respond, mean (SD), s                89.94 (64.89)       37.00 (35.31)
False positive rate, mean (SD), %            22.92 (38.70)         2.83 (6.63)
Attacks contained (%)                              75.3                75.0
----------------------------------------------------------------------------
MTTD   baseline minus AnDet Pro: t(392.3) = 3.10, p = 0.002, d = 0.28 (small)
MTTR   baseline minus AnDet Pro: t(391.3) = 11.27, p < 0.001, d = 1.01 (large)
FPR    baseline minus AnDet Pro: t(316.5) = 8.86, p < 0.001, d = 0.72 (medium)
Containment, two-proportion test: z = -0.09, p = 0.925
----------------------------------------------------------------------------
Requirements: 3 met, 3 missed, 4 not measured
One window scored in 25.4 ms (median); target 50 ms
Real test stream: AUROC AnDet Pro 0.147, baseline 0.918; normal windows flagged at the threshold: 0.5%
Trials in which at least half of the normal windows were flagged: AnDet Pro 0.0%, baseline 19.7%

Total time: 38.6 minutes

WHAT TO DO NEXT
  1. Open  /content/drive/MyDrive/AnDet/results/20261029_021641_standard/RESULTS_FOR_THESIS.md   (tables ar
  2. Open  /content/drive/MyDrive/AnDet/results/20261029_021641_standard/DOCUMENT_UPDATE_SHEET.md   (every
  3. Charts are in  /content/drive/MyDrive/AnDet/results/20261029_021641_standard/figures   and every table
```

Step 5: If Colab disconnected during the real run

1. Run Step 1 and Step 2 again.
2. Open Google Drive, go to AnDet, then results, and copy the name of the newest folder (for example `20261029_021500_standard`).
3. Paste that name into the cell below and run it.

Models that already finished training are loaded, so they are not trained again.

```
RESUME = 'PUT_THE_FOLDER_NAME_HERE'   # for example 20261029_021500_standard
!PYTHONUNBUFFERED=1 python run.py --mode standard --dataset {DATA_LOCAL} --results-dir {RESULTS} --set mode
```

Step 6: Get the results

The results are already saved in Google Drive, in AnDet, then results, then the newest folder. This cell also packs the newest run into one zip file and downloads it to your computer.

Open these two files first:

- `RESULTS_FOR_THESIS.md` has the tables and draft sentences for Chapter Four.
- `DOCUMENT_UPDATE_SHEET.md` lists every place in the thesis and the slides that must change.

Check that the first lines of `RESULTS_FOR_THESIS.md` say `Mode: standard`.

```
import glob
latest = open(f'{RESULTS}/LATEST.txt').read().strip()
print('Latest run:', latest)
shutil.make_archive('/content/andet_results', 'zip', latest)
from google.colab import files
files.download('/content/andet_results.zip')
```

```
Latest run: /content/drive/MyDrive/AnDet/results/20261029_021641_standard
```

Optional: run the automated tests

This takes about two minutes. At the end it should say `158 passed`.

```
!python -m pytest tests -q
```

```
........................................................................ [ 45%]
........................................................................ [ 91%]
..............                                                           [100%]
============================== warnings summary ===============================
tests/test_end_to_end.py::test_full_mode_runs_every_extra_check
  /content/AnDet_Pro/andet/analysis.py:448: RuntimeWarning: Mean of empty slice
    change_pct=_pct(np.nanmean(xa), np.nanmean(xb)))

tests/test_end_to_end.py::test_full_mode_runs_every_extra_check
  /content/AnDet_Pro/andet/analysis.py:461: RuntimeWarning: Mean of empty slice
    change_pct=_pct(np.nanmean(xa), np.nanmean(xb)))

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
158 passed, 2 warnings in 113.15s (0:01:53)
```